MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE VA-220B

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, tunable, waveguide output

ABSOLUTE RATINGS: See note 1

Parameter:	Ef	Ers	Er	F	Body temp	Ehk
Unit:	V	Vdc	Vdc	GHz	°C	Vdc
Maximum:	7.0	775	-1,000	7.425	150	±45
Minimum:	5.7	---	(See note 2)	7.125	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1

TEST CONDITIONS: See note 3

Parameter:	Ef	Ers	Er	F	Body temp
Unit:	V	Vdc	Vdc	GHz	°C
	6.3	750	-250 to -400	(See tunable freq)	100 (max)

GENERAL:

Qualification - Required

VA-220B

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1					
1231	Emission	Er = -350 to -450 Vdc; Ef = 5.7 V See notes 6 and 7	ΔIrs/ Irs	---	-15	%
4212	Resonator current	Er/max Po	Irs	---	80	mAdc
4213	Reflector voltage	Er/max Po, See note 2	Er	-250	-400	Vdc
4229	Total reflector current	Er = -350 to -450 Vdc See note 7	Ir	---	5	μAdc
1301	Heater current	Ef = 6.3 V	If	---	0.9	A
4223	Mechanical tuning	Er/max Po See note 9	F	7.425	7.125	GHz
4211	Power output	Er/max Po	Po	1.0	---	W
4201	Electronic tuning range	Er/50% max Po	Δ F	28	---	MHz
4231	Hysteresis	Er/max Po See note 8	Ratio	---	50	%
4223	Operating torque		Torque	---	32	oz-in.
	Quality conformance conformance, part 2					
----	Heater frequency modulation	Ef = 6.3 V at 400 Hz	Δ F	---	30	kHz (p-p)
4027	Temperature coefficient	Er/max Po: TA = +25° to +80°C See note 4	ΔF/ΔTA	---	±100	kHz/ °C
4215	Reflector voltage modulation coefficient	Er/max Po; Δ F < 5 MHz	ΔF/ΔEr	225	525	kHz/V
----	Beam voltage modulation coefficient	Er/max Po; ΔErs = +9 Vdc	ΔF/ΔErs	---	±150	kHz/V

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
----	Service-life guarantee	Er/max Po	t	7,500	---	hr
----	Service-life guarantee end point	Er/max Po	Po	0.80	---	W
	Periodic-Check Tests	See note 10				
----	Effective position of resonator	No voltages See note 5	---	0.106	0.156	in.

NOTES:

1. Ratings should not be exceeded under continuous or transient conditions. A single rating may be the limitation and simultaneous operation at another rating may not be possible. Design values for systems should include a safety factor to maintain operation within ratings under voltage and environmental variations.

2. Reflector voltage must always be negative with respect to the cathode by at least 20 volts.

3. All oscillation tests shall be made with the tube rigidly connected to a UG-344/U (MIL-F-3922/55) flange on RG-50/U (MIL-W-85/1) waveguide equipment. The load VSWR for the tube shall be less than 1.1:1. Forced-air cooling of at least 30 cfm is required in the fin and the waveguide as shown in figure 1. For maximum tube life, the tube temperature should be less than 100°C.

4. The temperature coefficient, when tested under the conditions specified, shall be computed in kilohertz per degree celsius over the temperature range specified.

5. The tube under test shall be adjusted to its center (resonance) frequency. Tune the signal source -150 MHz from the tube's center frequency and locate the position of the null on a standing wave indicator. Repeat the procedure with the signal source tuned to +150 MHz from the tube's center frequency. Replace the tube under test with a shorting plate, tune the signal source to the tube's center frequency and locate the position of the null on the standing-wave indicator. The effective resonator position is then defined as the position of the VSWR null with the shorting plate attached to the wave guide minus the average location of the VSWR null at ±150 MHz from the tube's center frequency with the tube attached to the waveguide. The VS-1120 adapter, or equal, may be used to change the effective resonator position of the tube 0.710 ±0.005 inches.

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NOTES: -Continued

6. The beam current shall not drop more than the specified percentage when the heater voltage is decreased from the normal operating potential to the lower specified value.

7. The test shall be made after a minimum warmup of 2 minutes. The tube shall not be oscillating during the test.

8. There shall be no power discontinuities between the half-power points for either direction of reflector voltage change through the tuning range specified under tunable frequency.

9. The tube shall be capable of being tuned smoothly and without discontinuities over a frequency range including the limits specified.

10. If the periodic-check tests have previously been performed by the tube manufacturer, and if materials, manufacturing processes and test requirements are the same as when the tests were previously performed, the periodic-check test(s) need not be performed.

Custodians:
Army - EL
Navy - EC
Air Force - 80

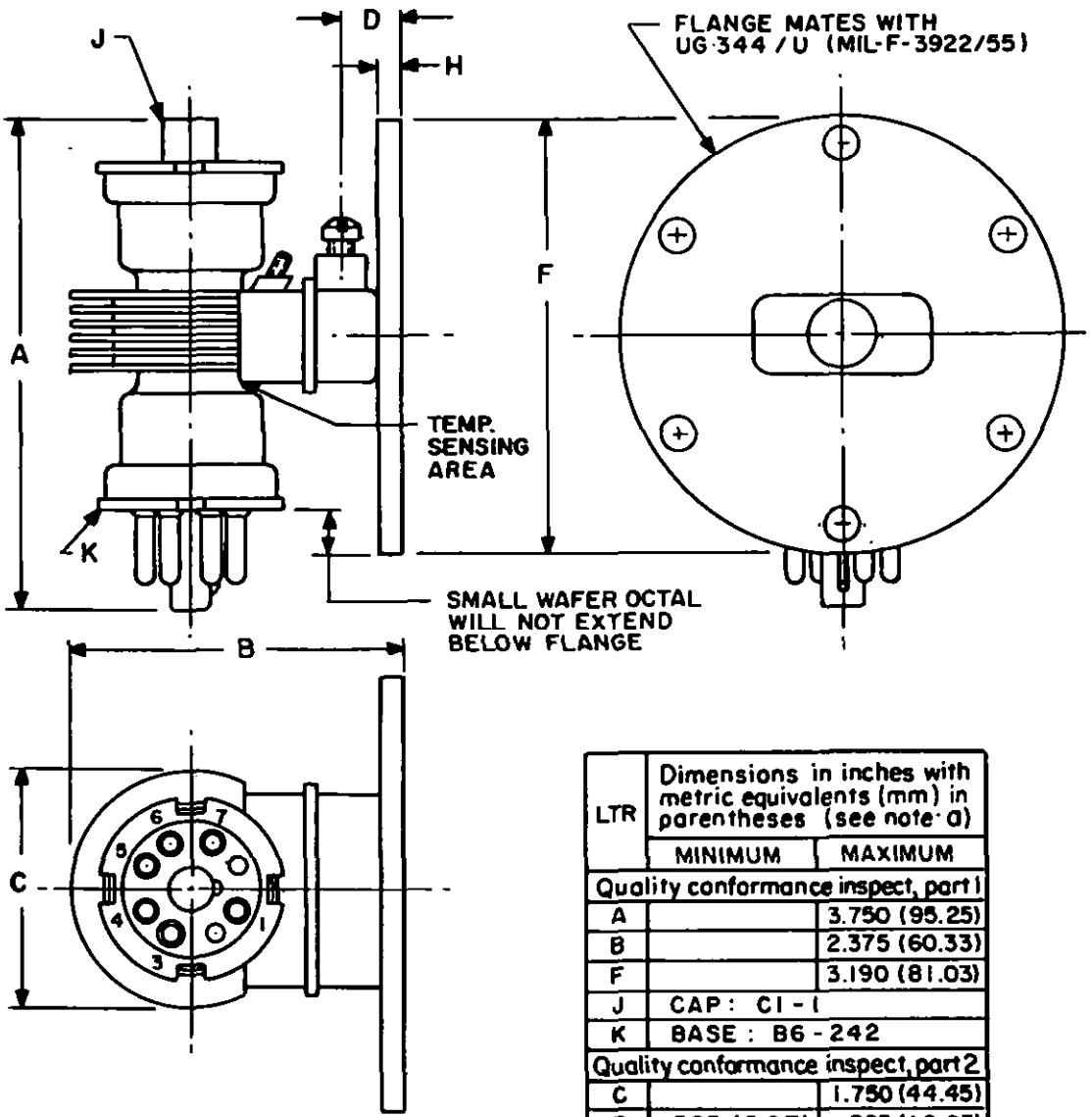
Preparing activity:
Air Force - 80

Review activities:
Army - EL, NSA

(Project 5960-2734)

User activities:
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Navy - MC, CG, AS, OS, SH
Air Force - 11

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LTR	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	MINIMUM	MAXIMUM
Quality conformance inspect, part 1		
A		3.750 (95.25)
B		2.375 (60.33)
F		3.190 (81.03)
J	CAP: C1-1	
K	BASE: B6-242	
Quality conformance inspect, part 2		
C		1.750 (44.45)
D	.365 (9.27)	.395 (10.03)
H	.105 (2.67)	.135 (3.43)

PIN CONNECTIONS	
PIN 1	SHELL
PIN 3	NO CONNECTION
PIN 4	CATHODE
PIN 5	HEATER
PIN 6	HEATER
PIN 7	NO CONNECTION
CAP	REFLECTOR

NOTE:

a. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type VA-220B

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